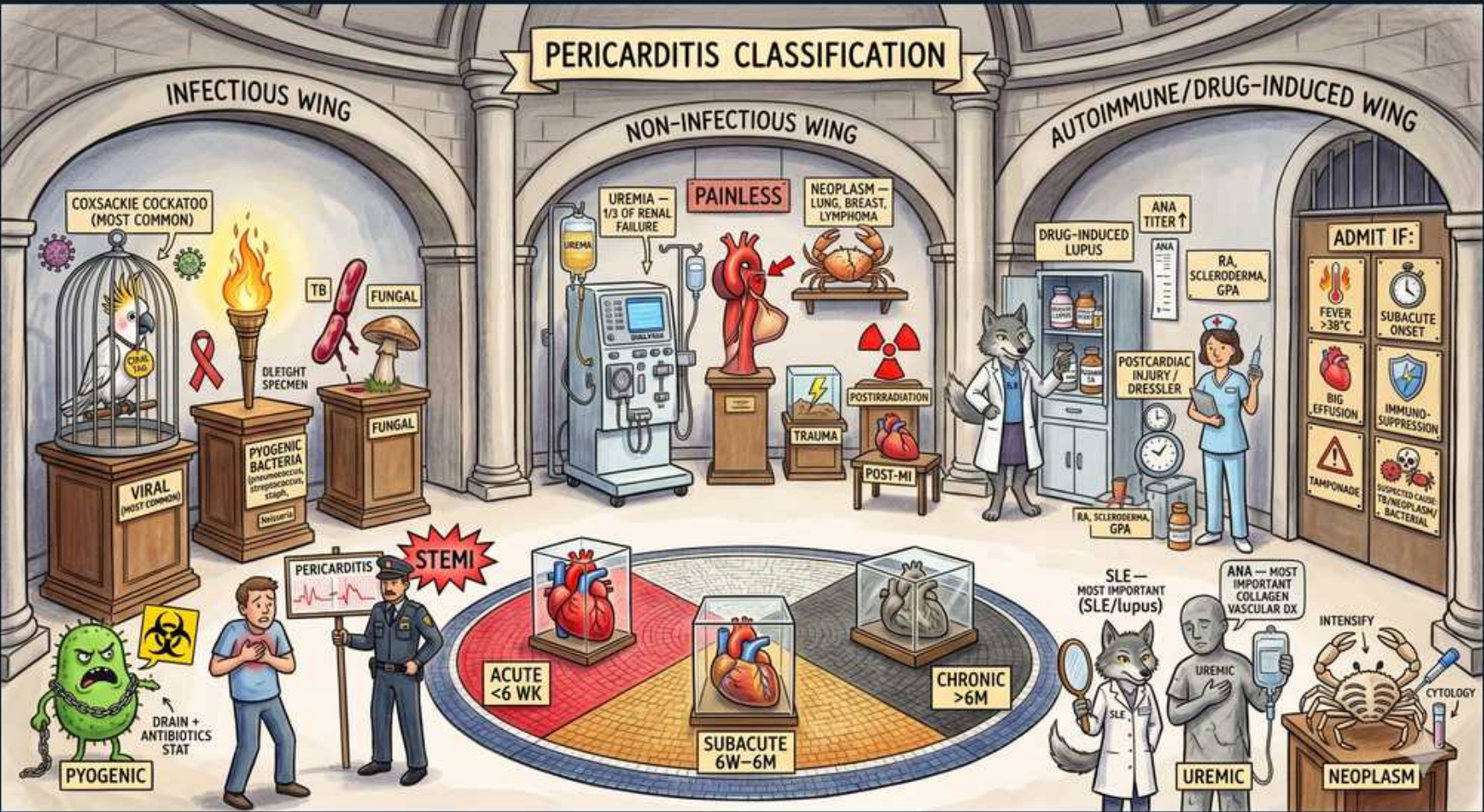


Pericardial Disease — Etiology & Classification





## 1. Classification museum

### WHAT YOU SEE

Banner 'PERICARDITIS CLASSIFICATION'

### WHAT IT ENCODES

Pericarditis is classified by ETIOLOGY (infectious vs non-infectious/autoimmune) and by TIME COURSE (acute <6 wk, subacute 6 wk–6 mo, chronic >6 mo). Diagnosis of acute pericarditis requires ≥2 of 4: pleuritic positional chest pain, pericardial friction rub, diffuse ST elevation/PR depression on ECG, and a new/worsening effusion. CRP/ESR are typically elevated and help track response.

## 2. Infectious wing

### WHAT YOU SEE

'INFECTIOUS WING' archway

### WHAT IT ENCODES

Infectious causes: viral (most common in developed world), bacterial/pyogenic (rare but lethal), fungal (endemic — histoplasma; or immunocompromised), and tuberculous (leading cause worldwide and a top cause of constriction). Infectious pericarditis is more often PAINFUL (vs the often-painless non-infectious causes).

## 3. Viral #1 (Coxsackie)

### WHAT YOU SEE

'COXSACKIE COCKATOO (MOST COMMON)' viral exhibit

### WHAT IT ENCODES

Viral pericarditis (classically Coxsackie B, also echovirus, adenovirus, EBV, influenza, HIV) is the most common identifiable infectious cause and usually follows a URI by 1–3 weeks. It is self-limited; treat with NSAIDs/aspirin + colchicine. Most 'idiopathic' cases are presumed viral. Watch troponin for myopericarditis.

## 4. Pyogenic → drain+abx

### WHAT YOU SEE

Green germ 'PYOGENIC — DRAIN + ANTIBIOTICS STAT'

### WHAT IT ENCODES

Purulent (bacterial) pericarditis — Staphylococcus, Streptococcus pneumoniae, Gram-negatives — is a medical emergency with high mortality if untreated. It presents with high fever, toxicity, and frequently tamponade. Management is URGENT pericardial drainage (often surgical, because pus is loculated/thick) PLUS IV antibiotics; antibiotics alone are insufficient.

### BOARD TRAP

WRONG to treat purulent pericarditis with antibiotics alone and observe. The discriminator: a febrile, toxic patient with thick/purulent fluid and rapid tamponade needs emergent DRAINAGE (pericardiocentesis or surgical window) in addition to IV antibiotics — delayed drainage risks death and constriction.

## 5. TB cause

### WHAT YOU SEE

'TB' red ribbon exhibit in infectious wing

### WHAT IT ENCODES

Tuberculous pericarditis is the leading cause of pericardial disease worldwide (especially with HIV) and the most common cause of constrictive pericarditis globally. It is subacute, with weight loss/night sweats and a serosanguinous, lymphocyte-rich effusion; diagnose with pericardial fluid ADA (elevated), AFB/culture, and biopsy. Treat with anti-TB therapy; adjunctive steroids may reduce constriction.

### BOARD TRAP

WRONG to forget TB in a chronic serosanguinous effusion just because the chest CXR is clear. The discriminator: elevated pericardial fluid ADA and lymphocytic exudate point to TB even without pulmonary findings; missing it leads to constrictive pericarditis.

## 6. Non-infectious wing

### WHAT YOU SEE

'NON-INFECTIOUS WING' archway, 'PAINLESS'

### WHAT IT ENCODES

Non-infectious causes are frequently PAINLESS and lack the classic ECG changes: uremia, malignancy, post-MI, radiation, hypothyroidism. Because they often don't hurt, they are detected via effusion, friction rub, or incidental imaging. This 'painless' theme is a high-yield board hook.

## 7. Uremic pericarditis

### WHAT YOU SEE

'UREMIA / END-RENAL FAILURE' exhibit

### WHAT IT ENCODES

Uremic pericarditis occurs with BUN typically >60 mg/dL in advanced/untreated renal failure; the inflammation is on the parietal surface only, so the ECG often shows NO diffuse ST elevation (unlike most other pericarditides). First-line treatment is intensified/urgent DIALYSIS, not NSAIDs/steroids. A related entity, dialysis-associated pericarditis, occurs in patients already on adequate dialysis.

### BOARD TRAP

Uremic pericarditis frequently shows NO diffuse ST elevation. WRONG to exclude pericarditis because the ECG lacks the classic changes, and WRONG to treat with anti-inflammatories first. The discriminator: a friction rub/effusion in a uremic patient with a 'normal-looking' ECG is uremic pericarditis — treat with intensified dialysis.

## 8. Neoplasm / malignancy

### WHAT YOU SEE

'NEOPLASM — LUNG, BREAST, LYMPHOMA' exhibit

### WHAT IT ENCODES

Malignant (metastatic) pericarditis from lung, breast, lymphoma, leukemia, and melanoma is a common cause of large/bloody and recurrent effusions and of malignant tamponade. Primary pericardial tumors (mesothelioma) are rare. Diagnose with fluid cytology; treat the effusion with a pericardial window/sclerotherapy plus therapy directed at the cancer.



## 9. Radiation

### WHAT YOU SEE

Radiation symbol exhibit, 'POST-RADIATION'

### WHAT IT ENCODES

Mediastinal radiation (e.g., for Hodgkin lymphoma or breast cancer) causes acute pericarditis early and, years later, fibrosis leading to constrictive pericarditis. Radiation constriction carries a notably worse pericardiectomy outcome because the myocardium and coronaries are often co-injured.

### BOARD TRAP

WRONG to assume pericardiectomy will fix radiation-induced constriction as readily as idiopathic constriction. The discriminator: prior chest radiation implies concurrent myocardial/coronary damage, so surgical outcomes are poorer and the picture may overlap with restrictive cardiomyopathy.

## 10. Post-MI / Dressler

### WHAT YOU SEE

'POST-MI' note in non-infectious wing

### WHAT IT ENCODES

Two distinct post-MI entities: (1) early peri-infarction (epistenocardiac) pericarditis within 1–3 days, from direct transmural inflammation; (2) Dressler syndrome 2–8 weeks later — an immune-mediated pericarditis with fever, pleuritis, and elevated ESR. Treat both with aspirin ± colchicine; avoid non-aspirin NSAIDs and steroids early after MI (impair healing).

### BOARD TRAP

WRONG to confuse early peri-infarction pericarditis (days) with Dressler (weeks, autoimmune). The discriminator is timing and mechanism. Also WRONG to treat either with ibuprofen/indomethacin or steroids in the early post-MI window — these impair scar healing and raise rupture risk; use aspirin ± colchicine.

## 11. Autoimmune/drug wing

### WHAT YOU SEE

'AUTOIMMUNE / DRUG-INDUCED WING', SLE, RA, drug-induced lupus

### WHAT IT ENCODES

Autoimmune/connective-tissue causes: SLE (most important), rheumatoid arthritis, systemic sclerosis, Sjögren, and drug-induced lupus (procainamide, hydralazine, isoniazid, minocycline — anti-histone antibodies). Pericarditis can be the presenting serositis of SLE. Treat the underlying disease ± NSAIDs/colchicine; steroids for the autoimmune flare.

## 12. SLE most important

### WHAT YOU SEE

'SLE — MOST IMPORTANT COLLAGEN VASCULAR DISEASE'

### WHAT IT ENCODES

SLE is the most important connective-tissue cause of pericarditis; pericarditis is the most common cardiac manifestation of lupus and a feature of acute serositis. The effusion is typically an exudate with low complement and positive ANA/anti-dsDNA. Manage with NSAIDs/colchicine for mild disease and corticosteroids/immunosuppression for the lupus flare.

### BOARD TRAP

WRONG to manage lupus pericarditis with NSAIDs/colchicine alone when it reflects an active SLE flare. The discriminator: low serum complement, high anti-dsDNA, and other organ involvement indicate a flare needing corticosteroids/immunosuppression, not just anti-inflammatory symptom control.

## 13. Acute <6wk

### WHAT YOU SEE

Time-course dais 'ACUTE <6 WK'

### WHAT IT ENCODES

Time course: ACUTE <6 weeks; SUBACUTE 6 weeks–6 months; CHRONIC >6 months. Most acute pericarditis is viral/idiopathic and self-limited. High-risk features predicting complications and warranting admission: high fever >38°C, subacute onset, large effusion/tamponade, immunosuppression, oral anticoagulation, trauma, and failure to respond to NSAIDs within a week.

## 14. Chronic >6mo

### WHAT YOU SEE

'CHRONIC >6M' statue on the dais

### WHAT IT ENCODES

Chronic pericarditis (>6 months) encompasses chronic effusion, chronic constrictive pericarditis, and recurrent/incessant pericarditis. Chronic and recurrent disease (especially after inadequate initial colchicine) is the bridge to constriction; etiologies skew toward TB, radiation, neoplasm, and autoimmune.